

Lab program-20:

20. Construct a C program to simulate Reader-Writer problem using semaphores

Code:

```
#include <stdio.h>
#include <windows.h>

HANDLE mutex, writeBlock;

int data = 0;
int readersCount = 0;

DWORD WINAPI reader(LPVOID arg)
{
    int i = 0;

    while (i < 10)
    {
        // Entry Section
        WaitForSingleObject(mutex, INFINITE);

        readersCount++;

        if (readersCount == 1)
        {
            WaitForSingleObject(writeBlock, INFINITE);
        }

        ReleaseSemaphore(mutex, 1, NULL);

        // Reading
        printf("Reader reads data: %d\n", data);

        // Exit Section
        WaitForSingleObject(mutex, INFINITE);

        readersCount--;

        if (readersCount == 0)
        {
            ReleaseSemaphore(writeBlock, 1, NULL);
        }

        ReleaseSemaphore(mutex, 1, NULL);
    }
}
```

```

        Sleep(500);

        i++;
    }

    return 0;
}

DWORD WINAPI writer(LPVOID arg)
{
    int i = 0;

    while (i < 10)
    {
        WaitForSingleObject(writeBlock, INFINITE);

        // Writing
        data++;

        printf("Writer writes data: %d\n", data);

        ReleaseSemaphore(writeBlock, 1, NULL);

        Sleep(1000);

        i++;
    }

    return 0;
}

int main()
{
    HANDLE readerThread;
    HANDLE writerThread;

    // Create semaphores
    mutex = CreateSemaphore(NULL, 1, 1, NULL);
    writeBlock = CreateSemaphore(NULL, 1, 1, NULL);

    if (mutex == NULL || writeBlock == NULL)
    {
        printf("Failed to create semaphores.\n");
        return 1;
    }

    // Create threads
    readerThread = CreateThread(

```

```
        NULL,  
        0,  
        reader,  
        NULL,  
        0,  
        NULL  
    );  
  
    writerThread = CreateThread(  
        NULL,  
        0,  
        writer,  
        NULL,  
        0,  
        NULL  
    );  
  
    // Wait for threads  
    WaitForSingleObject(readerThread, INFINITE);  
    WaitForSingleObject(writerThread, INFINITE);  
  
    // Cleanup  
    CloseHandle(readerThread);  
    CloseHandle(writerThread);  
    CloseHandle(mutex);  
    CloseHandle(writeBlock);  
  
    return 0;  
}
```

OUTPUT:

```
PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_20.exe
Reader reads data: 0
Writer writes data: 1
Reader reads data: 1
Writer writes data: 2
Reader reads data: 2
Reader reads data: 2
Writer writes data: 3
Reader reads data: 3
Reader reads data: 3
Writer writes data: 4
Reader reads data: 4
Reader reads data: 4
Writer writes data: 5
Reader reads data: 5
Reader reads data: 5
Writer writes data: 6
Writer writes data: 7
Writer writes data: 8
Writer writes data: 9
```